Muscle weakness consequences, compensation and assistance

Project goals

The **goals** of this project are to analyze how muscles' force production and the corresponding effort/cost during normal walking are affected by decreased muscle strength, by increased synergist muscle strength, and by a passive assistive device. **Part 1** introduces the problem – a weak muscle, and parts 2 and 3 introduce two ways of compensating for the weak muscle – by strengthening an agonist muscle (**Part 2**) and with an external assistive exoskeleton (**Part 3**).

Let us refer here to **muscle forces as the resulting "force production" to perform a task/provide joint moment**, and to the **muscle activations as the "effort" or "cost"** of producing this joint moment.

Optimizations to determine force distribution in net joint moments often have the objective to minimize metabolic effort, usually determined as a function of muscle activations which reflect metabolic cost to some degree. By comparing muscle activations for given net joint moments during a movement, one can analyze the cost and the efficiency of a movement.

Baseline force production and cost are considered as those computed with default 100% muscle strength. **For all analyses, use the model and files we provide with this assignment, not the ones from previous projects.** We analyze the **left** gait cycle (5.11s to 6.09s).

For all parts: Do <u>not</u> plot lots of raw results from all muscles – choose and illustrate findings from the most relevant muscles and/or the ones that change the most, or generally in an overall meaningful and clear way. Make an effort to illustrate your results well – you won't be able to spend a lot of time explaining them. Grades will be reduced for unclear plots/figures, illustrations with illegible text, lacking correct units and clear descriptions, etc.

Part 1 (worth up to 6 points):

Study question

Can a person with weakness in one of the plantarflexor muscles achieve normal walking by compensating with other plantarflexor muscles? If so, how, and at what cost?

Tasks

- Load the provided gait model with static optimization (SO) setting files. Muscle forces and activations should be computed through SO.
- Simulate muscle weakness in the **left soleus** muscle. Reduce strength to 0%, in at least 3 steps (for instance, reduce by 33%, then by 66% then by 100%).

Discuss

- <u>Analyze</u> how activations and forces in this muscle and in other muscles change and <u>interpret</u> according to the project goals. Elaborate.
- Based on evaluation of the model outputs, conclude whether the model still has the capacity to walk normally at each strength reduction level. How specifically did you come to this conclusion? *Recall that OpenSim has elements that make it possible to find a solution, but that might not necessarily mean that the muscles are able to generate the required joint moment.*

- Describe how and why the forces in the left soleus, in its agonists, its antagonists and even in more proximal muscles are influenced by weakness in the soleus. Recall that muscles often have several functions depending on whether they are uni- or bi/multiarticular, and that different muscles are important in different phases of gait.
- What are the consequences in terms of **cost**, both locally at the joint and overall, for each level of strength reduction?

Part 2 (Worth up to 6 points)

Study question

For a person with weakness in the soleus, what influence does increased strength in an agonist muscle have? Can it compensate for weakness in the soleus and restore balance in agonists, antagonists and distant muscles? Fully or only partially? Is it an efficient solution? As efficient as the baseline condition?

Tasks

- Reduce muscle strength in the left soleus muscle. You can choose the level of strength reduction, for instance at 50% or 33% strength as in Part 1. Maintain this strength level in the left soleus for Part 2.
- Simulate *increased* strength in the gastrocnemius muscle, an **agonist** to the left soleus muscle, by a few increments. Treat the medial and lateral gastrocnemius heads at the same time – if you reduce/increase strength by, say, 25%, then do this to both at the same time.

Discuss

- How did this compensation affect the muscle activity in the soleus and gastrocnemius muscle? Elaborate.
- With each new synergistic pair, does the model have the capacity to perform the movement? How did you conclude this? Elaborate.
- How does this new synergistic pair influence muscle forces in their synergist muscle(s), antagonist muscle(s) as well as other lower limb muscles?
- Does the cost change?
- Is this an efficient solution, based on 1) Baseline and 2) weakness in the soleus and default strength in the gastrocnemius? Elaborate, based on the study question and on the project goals.
- Would you consider this simulation to be a realistic concept for a therapy suggestion? Why or why not?

Part 3 (worth up to 8 points)

Study question

Given a large amount of weakness in the soleus muscle, can a passive exoskeleton that consists of a unidirectional (extension) linear spring (recall Project 3) compensate for this weakness? How could it be implemented? Is this an efficient solution? Can it be comfortable?

Tasks

- Reduce strength in the left soleus muscle by a large amount but not entirely, for instance strength reduced to 25% (you choose). Return the strength in the gastrocnemius muscle to its baseline/100% strength value.
- Add a **uniarticular**, unidirectional linear spring (elastic band) as you did in Project 3, starting with what you identified as optimal stiffness, and optimal and realistic resting length and attachment. This is to

approximately represent a spring-based passive ankle exoskeleton. Feel free to modify spring parameters and attachment points based on the model and on the aims of this project. Note that it is probably necessary to move both attachment points away from the bone surfaces.

- With this passive exoskeleton, perform SO again and compute individual muscle activations and forces.

Discuss

- Make a reasonable sketch of an actual device that includes the user and the spring's geometry (a hand-drawn sketch is fine). Place this sketch side-by-side with the OpenSim model of the spring. Try to be clear on how it could actually be worn by the user – how and where it would be attached. I suggest that you also illustrate the exoskeleton and spring in 2 or 3 different positions, for instance at a neutral ankle angle and at a dorsiflexed angle.
- Plot the spring force and its moment contribution, with respect to the overall joint moment. How much of the net joint moment can the exoskeleton account for? Comment on the magnitude of the spring force.
- Can the device compensate for weakness of the soleus muscle? Completely or partially? What specific criteria and/or findings do you use when you conclude this? Elaborate. Optional challenge: modify the device parameters to try to achieve lower than baseline overall cost.
- Is this an efficient solution, compared to 1) Baseline and 2) weak soleus muscle but no exoskeleton (Part 1)? Elaborate and motivate your conclusions with relevant results/findings.
- The device must transmit the spring force to the user's body segments in order for it to contribute to ankle plantarflexion moment. Sketch the positions and approximate directions and magnitudes of the *spring forces* and of the *interaction forces* of the device onto the user – separate illustrations for different body segments and for the device. Estimate the interaction forces' direction, magnitude and location(s). Indicate where the risks for discomfort/pain are. Do you have any suggestions for comfort?
- Given what you know now after this course, make some suggestions or speculations for a better/more efficient/more comfortable intervention for weakness in the soleus muscle, conceptually – it may be a spring-based ankle exoskeleton but it need not be, it can be anything.

Format

No reports this time! Each group will present their findings to Lanie and Jennifer orally in January. You must, however, hand in either a soft copy or a hard copy of your presentation/slides (in color) no later than a half working day before your oral presentation.

Presentations should be **a maximum of 20 min**, which will be followed by at least 20 minutes of questions. You can book your presentation time on Canvas.

Note

- *Every group member* should be able to answer at least general questions about *all* parts of the project.
- We strongly suggest that you present in person if you are able. You may present on zoom or hybrid if some in the group are unable to attend in person.
- Please number your slides
- Please write *both* the group number and all names on the title slide.